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Interfacial bonding layer for an anode-free solid-state-battery

Abstract

An anode-free solid-state battery may include a cathode layer; an anode current collector layer; and a separator layer between the cathode layer and the anode current collector layer. The battery can further include an anti-dendrite layer located between the separator layer and the anode current collector layer. The battery further includes an interfacial bonding layer located between the anti-dendrite layer and the anode current collector layer. The interfacial bonding layer increases an amount of electrical connectivity between the anode current collector layer. A first amount of adhesion between the interfacial bonding layer and the anode current collector layer can be greater than a second amount of adhesion between the anti-dendrite layer and the interfacial bonding layer.

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Background/Summary

BACKGROUND

(1) During charging of a lithium ion battery, lithium ions migrate from battery's cathode to the battery's anode through a separator located between the cathode and anode. Through a process called intercalation, lithium ions become inserted into the material functioning as the anode. During this charging process, lithium ions may also plate onto a surface of the anode facing the separator. The nucleation energy associated with the anode may encourage lithium ions to plate on top of other lithium that has already plated onto the anode (rather than plating in a roughly even film across the surface of the anode). These pools of lithium can form dendrites. A dendrite can be a protrusion of lithium metal that extends away from the surface of the anode towards the separator. Over time (e.g., through multiple charge and discharge cycles), dendrites may grow to such a length that the dendrite pierces the separator and directly electrically connects (i.e. short circuits) the cathode with the anode. Such an electrical connection can result in destruction of the battery and possibly serious side-effects, such as overheating and fire.

SUMMARY

(2) Various embodiments are described related to an anode-free solid-state battery. In some embodiments, an anode-free solid-state battery is described. The battery may comprise a cathode layer. The battery may comprise an anode current collector layer. The battery may comprise a separator layer between the cathode layer and the anode current collector layer. The battery may comprise an anti-dendrite layer located between the separator layer and the anode current collector layer. The anti-dendrite layer may discourage dendrite formation. The battery may comprise an interfacial bonding layer located between the anti-dendrite layer and the anode current collector layer. The interfacial bonding layer may increase an amount of electrical connectivity between the anode current collector layer and lithium deposited on the interfacial bonding layer during charging of the anode-free solid-state battery. A first amount of adhesion between the interfacial bonding layer and the anode current collector layer may be greater than a second amount of adhesion between the anti-dendrite layer and the interfacial bonding layer.

(3) Embodiments of such a method may include one or more of the following features: a presence of the interfacial bonding layer may decrease an impedance of the anode-free solid-state battery. The interfacial bonding layer may comprise acetylene black. The anti-dendrite layer may comprise one or more materials selected from the group consisting of: carbon black; acetylene black; ketchen black; silver; zinc; gold; bismuth; tin; polyvinylidene fluoride (PVDF); polyimide (PI); polyacrylic acid (PAA); and carboxymethyl cellulose styrene-butadiene rubber (CMC-SBR). The separator layer may comprise: a scaffolding material; a salt; a solvent; and an additive. The salt and additive may be dissolved into the solvent to form a liquid electrolyte. The liquid electrolyte may be dispersed throughout the scaffolding material. The liquid electrolyte, after being dispersed throughout the scaffolding material may be converted to a gel. The interfacial bonding layer may be between 0.05 μm and 5 μm . The anti-dendrite layer may be between 0.05 micrometers and 10 micrometers in thickness.

(4) In some embodiments, a method for creating an anode-free solid-state battery is described. The method may comprise depositing an interfacial bonding layer onto an anode current collector layer. The method may comprise depositing an anti-dendrite layer onto a lithium-gel separator layer. The method may comprise layering a cathode and cathode current collector onto the lithium-gel separator layer. The method may comprise layering the anti-dendrite layer onto the interfacial bonding layer. A first amount of adhesion between the anti-dendrite layer and the interfacial bonding layer may be less than a second amount of adhesion between the interfacial bonding layer and the anode current collector layer. The interfacial bonding layer may increase an amount of electrical connectivity between the anode current collector layer and lithium deposited on the interfacial bonding layer.

(5) Embodiments of such a method may include one or more of the following features: a presence of the interfacial bonding layer may decrease an impedance of the anode-free solid-state battery. The interfacial bonding layer may comprise acetylene black. The method may further comprise creating the lithium-gel separator layer. Creating the lithium-gel separator layer may comprise attaching a first adhesive layer to scaffolding material. The scaffolding material may define voids within the scaffolding material. The method may comprise attaching a second adhesive layer to the scaffolding material. The method may comprise permeating a liquid electrolyte into the voids within the scaffolding material. The anti-dendrite layer may be deposited onto the lithium-gel separator layer prior to the liquid electrolyte being permeated into the voids within the scaffolding material. Creating the lithium-gel separator layer may further comprise, after permeating the liquid electrolyte into the voids within the scaffolding material, applying heat to the lithium-gel separator layer that may cause the liquid electrolyte to become a gel electrolyte. The interfacial bonding layer may comprise a binder material and a conductive material. The binder material may be PVDF and the conductive material may be ketchen black. The anti-dendrite layer may comprise one or more materials selected from the group consisting of: carbon black; acetylene black; ketchen black;

silver; zinc; gold; bismuth; tin; polyvinylidene fluoride (PVDF); polyimide (PI); polyacrylic acid (PAA); and carboxymethyl cellulose styrene-butadiene rubber (CMC-SBR).

(6) In some embodiments, a pouch-style anode-free solid-state battery is described. The battery may comprise a flexible pouch. The battery may comprise a cathode layer. The battery may comprise an anode current collector layer. The battery may comprise a separator layer between the cathode layer and the anode current collector layer. The battery may comprise an anti-dendrite layer located between the separator layer and the anode current collector layer. The anti-dendrite layer may discourage dendrite formation. The battery may comprise an interfacial bonding layer located between the anti-dendrite layer and the anode current collector layer. The interfacial bonding layer may increase an amount of electrical connectivity between the anode current collector layer and lithium deposited on the interfacial bonding layer. A first amount of adhesion between the interfacial bonding layer the anode current collector layer may be greater than a second amount of adhesion between the anti-dendrite layer and the interfacial bonding layer. The cathode layer, the anode current collector layer, the separator layer, the anti-dendrite layer, and the interfacial bonding layer may be inserted within the flexible pouch.

(7) Embodiments of such a device may include one or more of the following features: the battery may comprise a liquid electrolyte that becomes a gel when exposed to heat. The interfacial bonding layer may comprise acetylene black.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

(1) FIG. 1 illustrates an embodiment of a layer stack of an anode-free solid-state battery having a lithium gel separator layer and an anti-dendrite layer.

(2) FIG. 2 illustrates another embodiment of a layer stack of an anode-free solid-state battery having lithium gel separator layers and anti-dendrite layers.

(3) FIG. 3A illustrates an embodiment of a lithium gel separator layer being formed using heat.

(4) FIG. 3B illustrates an embodiment of a method for creating a lithium gel separator layer.

(5) FIG. 4 illustrates an embodiment of a method for manufacturing a pouch-style battery cell that contains an anode-free solid-state battery having a lithium gel separator layer and an anti-dendrite layer.

(6) FIG. 5 illustrates an embodiment of a cylindrical battery press system.

(7) FIGS. 6A and 6B illustrates an embodiment of a method for creating a cylindrical anti-dendrite anode-free solid-state battery.

(8) FIG. 7 illustrates an embodiment of a layer stack of an anode-free solid-state battery having a lithium gel separator layer, an anti-dendrite layer, and a interfacial bonding layer.

(9) FIG. 8 illustrates an embodiment of a layer stack of an anode-free solid-state battery that indicates relative amounts of adhesion between various layers.

(10) FIG. 9 illustrates an embodiment of a layer stack of an anode-free solid-state battery being charged in which lithium is being deposited onto the interfacial bonding layer.

(11) FIG. 10 illustrates an embodiment of a method for manufacturing a pouch-style battery cell that contains a solid-state battery having a lithium gel separator layer, an anti-dendrite layer, and an interfacial bonding layer.

DETAILED DESCRIPTION

(12) Introduction of an anti-dendrite layer in combination with a lithium gel separator layer can inhibit the growth of dendrites while not increasing the thickness of the battery cell by a large amount. An anti-dendrite layer may be coated directly onto an anode current collector of an anode-free solid state battery (SSB). In an anode-free SSB, the anode current collector, which can be a copper foil, may effectively function as both the anode and the anode current collector. The anti-

dendrite layer may decrease the nucleation energy needed for lithium ions to deposit as lithium metal onto the surface of the anode current collector that is in contact with the anti-dendrite layer. Rather than lithium ions tending to plate on top of lithium metal that has already plated onto the surface of the anode current collector (thus causing pools, which can lead to dendrite), lithium may tend to deposit in a roughly even film across the surface of the anode current collector.

(13) The anti-dendrite layer may be in direct contact with a lithium gel separator layer. The lithium gel separator layer may serve multiple functions. First, the lithium gel separator layer can function as a solid-state electrolyte to facilitate movement of lithium ions between the cathode and the anode. The lithium gel separator layer also serves as a separator to prevent a direct electrical connection between the cathode and anode. The lithium gel separator layer further can have characteristics that further inhibit dendrite growth.

(14) Further detail regarding such embodiments and additional embodiments is provided in relation to the figures. FIG. 1 illustrates an embodiment of a layer stack **100** of an anode-free solid-state battery having a lithium gel separator layer and an anti-dendrite layer. Layer stack **100** can include: cathode current collector **110**; cathode **120**; lithium gel separator layer **130**; anti-dendrite layer **140**; and anode current collector **150**.

(15) Cathode current collector **110** may be a conductive film that is layered with cathode **120**. Cathode current collector **110** may, for example, be an aluminum foil. Other forms of conductive foils are possible. Cathode **120** may, for example, be NCA (Nickel-Cobalt-Aluminum Oxide) or NCM (nickel-manganese-cobalt).

(16) Cathode **120** may have a first surface in direct contact with cathode current collector **110**, an opposite surface of cathode **120** can be in direct contact with lithium gel separator layer **130**. Lithium gel separator layer **130** can function as a solid-state electrolyte (in the form of a gel) to facilitate movement of lithium ions between cathode **120** and anode current collector **150**. Lithium gel separator layer **130** also serves as a separator to prevent a direct electrical connection between cathode **120** and anode current collector **150**. Lithium gel separator layer **130** can have characteristics that further inhibit dendrite growth. Lithium gel separator layer **130** may be initially at least partially in liquid form. After assembly, a process may be applied to convert the liquid to a gel form. Such a process may involve the application of pressure, heat, or both. Further detail regarding lithium gel separator layer **130** is provided in relation to FIG. 3A.

(17) Lithium gel separator layer **130** may have a first surface in direct contact with cathode **120**. A second surface opposite the first surface of lithium gel separator layer **130** may be in direct contact with anti-dendrite layer **140**. Anti-dendrite layer **140** may have several key characteristics: First, anti-dendrite layer **140** may decrease the nucleation energy necessary for lithium ions to plate as lithium metal on the surface of anode current collector **150** in direct contact with anti-dendrite layer **140**. By decreasing the nucleation energy, it may be more likely that lithium ions will deposit directly onto anode current collector **150** rather than “pooling” or depositing onto lithium metal that has already plated on anode current collector **150**.

(18) A second key characteristic of anti-dendrite layer **140** is that an amount of adhesion between anti-dendrite layer **140** and anode current collector **150** is less than an amount of adhesion between anti-dendrite layer **140** and lithium gel separator layer **130**. By adhesion being less between the surfaces of anti-dendrite layer **140** and anode current collector **150**, lithium is encouraged to plate between anti-dendrite layer **140** and anode current collector **150** as opposed to between lithium gel separator layer **130** and anti-dendrite layer **140**.

(19) Anti-dendrite layer **140** may be relatively thin. For instance, anti-dendrite layer **140** may be between 0.05 μm and 10 μm . In some embodiments, anti-dendrite layer **140** may be deposited as a film on a surface of anode current collector **150**. Anti-dendrite layer **140** may be made of one or more of the following materials: carbon black; acetylene back black; silver; zinc; gold; bismuth; tin; polyvinylidene fluoride (PVDF); polyimide (PI); polyacrylic acid (PAA); and carboxymethyl cellulose styrene-butadiene rubber (CMC-SBR). Anti-dendrite layer **140** may also be formed using

alloys of silver, zinc, gold, bismuth, and tin. While anti-dendrite layer **140** may be formed from a single type of material; multiple materials indicated may be used to form anti-dendrite layer **140**. (20) Anode current collector **150** may have a first surface in contact with anti-dendrite layer **140**. Anode current collector **150** can function as both the anode and the anode current collector. In some embodiments, anode current collector **150** is a conductive foil, such as a copper foil. Without anti-dendrite layer **140** being present, anode current collector **150** may exhibit a higher nucleation energy that tends to cause lithium to pool rather than deposit as a film during charging of the battery cell.

(21) FIG. 2 illustrates another embodiment of a layer stack **200** of an anode-free solid-state battery having lithium gel separator layers and anti-dendrite layers. In some embodiments, multiple sets of layers may be layered together to increase the charge capacity of the battery cell. In the illustrated embodiment of FIG. 2, layers **110** through **150** are as detailed in relation to FIG. 1. Additionally, anti-dendrite layer **210** is layered on an opposite side of anode current collector **150** from anti-dendrite layer **140**. A second lithium gel separator layer **202** is in direct contact with anti-dendrite layer **210**. Further, cathode **230** and cathode current collector **240** are layered against lithium gel separator layer **220**. Additional layers may be added in the same manner as detailed in relation to FIG. 2. For instance, below cathode current collector **240** may be another cathode, followed by another lithium gel separator layer, etc. For instance many sets of layers may be added to increase the charge capacity of the battery cell. While FIG. 1 shows a single stack set and FIG. 2 illustrates a double stack set; other embodiment may include may more stacks, such as 16 or more. Such layers, once together, may be sealed as part of a pouch-style battery cell.

(22) FIG. 3A illustrates an embodiment of a lithium gel separator layer being formed using heat. FIG. 3B, detailed herein in parallel with FIG. 3A, illustrates an embodiment of a method for creating a lithium gel separator layer. The lithium gel separator layer may function as a phase-changing electrolyte, which can be used in a solid state battery. A lithium gel separator layer, such as lithium gel separator layer **130**, can include multiple sublayers and may be created using heat. Pressure may also be used to increase the surface area of interfaces between layers of the lithium gel separator layer. First, a non-reactive scaffold may be formed at block **350**. For example, scaffolding material **305** may be polyethylene (PE) or polyethylene oxide (PEO). Scaffolding material **305** may be permeable such that a liquid, such as an electrolyte liquid can be permeated, injected, or otherwise introduced to scaffolding material **305**. The physical structure of scaffolding material **305** may create gaps that can be filled with liquid. For instance, scaffolding material **305** may have a porosity of between 10% and 90%, into which a liquid can be introduced. The specific physical structure may be honeycomb structure, spider-web structure, or some other pattern or random porous physical structure that allows liquid to fill empty spaces within scaffolding material **305**. The scaffolding layer may be between 1 μm and 100 μm thick. In some embodiments, the scaffolding layer is 15 μm thick.

(23) At block **355**, a first adhesive layer may be attached to the non-reactive scaffold. At block **360**, a second adhesive layer may be attached to an opposite side of the non-reactive scaffold. Therefore, scaffolding material **305** may be located between two adhesive layers **310**. Together, these three layers may form dry separator layer **300** in which an electrolyte has not yet been introduced. Each of adhesive layers **310** may include either PvDF, PI, PAA, or CMC-SBR. Such materials can function as an adhesive bonder. Therefore, adhesive layers **310** can serve to increase the amount of adhesion between scaffolding material **305**, cathode **120**, and anti-dendrite layer **140**. An amount of adhesion between lithium gel separator layer **130** and anti-dendrite layer **140**, at least in part due to an adhesion layer, may be greater than an amount of adhesion between anti-dendrite layer **140** and anode current collector **150**.

(24) In some embodiments, a ceramic may be added to one or both adhesive layers that improves lithium ion transport and can help discourage dendrite formation. Such ceramics can include: MgO, PZT, BaTiO₃, SBT, BFO, LATPO, LISICON, LICGC, LAGP, LLZO, LZO, LAGTP, LiBETI,

LiBOB, LiTf, LLTO, LLZP, LTASP, and LTZP. Each of adhesive layers **310** may be between 1-500 um in thickness. Using ceramics within the adhesive layers can decrease ionic-conductivity. Using a Li-ion conductor ceramic may still decrease ionic-conductivity (compared with liquid), but can secure higher ionic-conductivity compared with other ceramics. However, the advantage of a ceramic being able to prevent a short-circuit and to decrease overall cell failure rate may out-weigh the drawback of the decreased ionic conductivity.

(25) At block **365**, a liquid electrolyte mixture may be created. The liquid electrolyte mixture may include: a lithium salt; a solvent; and additives. The salt may be LiFSI, LiTFSI, or LiPF₆. The concentration of the salt may be between 1.0 to 4.0 mole per liter. The lithium salt may allow the lithium liquid to function as an electrolyte. The solvent may be: dimethyl carbonate (DMC), dimethoxyethane (DME), diethyl carbonate (DEC), dioxolane (DOL), bis trifluoroethyl ether (BTFE), ethyl methyl carbonate (EMC), or ethylene carbonate (EC). The solvent may function to dissolve the salt.

(26) The additives may include the compounds within the lithium liquid that causes a transition from a liquid to a gel when heat is applied. Generally, the additives comprise a polymer and cross-linker. When heat of, for example, between 60° C. and 150° C. is applied to the additive, the cross-linker ignites and causes further polymerization of the polymer and the solvent. Since the lithium salt is evenly distributed throughout the solvent, when the gel is formed, the lithium salt will be evenly distributed throughout the gel. The one or more additives may include CsPF₆, FEC (fluoroethylene carbonate), polycarbonate (PC), or LiNO₃. The additive may have a concentration of 0.01-4.0 moles per liter. The additives, includes the polymer additive and the cross-linker additive may be mixed into the lithium liquid before the lithium liquid is permeated throughout the non-reactive scaffolding.

(27) One possible combination of lithium salt and solvent may be 4 M of LiFSI dissolved in DME, to which the additives can be added. Table 1 indicates a combination of polymer additive, cross-linker additive, and the relative concentrations that can be used of the polymer additive and cross-linker additive.

(28) TABLE-US-00001 TABLE 1 Cross-Linker Polymer:Cross- Polymer Additive Additive Linker (g/L) P4VP (poly(4- C12TFSA 50:50 vinylpyridine)) P4VP C6TFSA 50:70 PDMEMA (poly(2- C12TFSA 50:50 dimethylamino- ethylmethacrylate)) P4VP C12TFSA 50:50 PDMEMA C12TFSA 50:50 PDMEMA C12TFSA 100:50 PDMEMA C12TFSA 40:50 P4VP C12TFSA 40:50

(29) In some embodiments, one or more additional additives may also function to reduce side reactions. A purpose of adding an additive can be to help form LiF, namely solid electrolyte interphase (SEI), which can prevent Li metal from having various side reactions.

(30) While the lithium gel separator layer is in the form of dry separator layer **300**, dry separator layer **300** may be layered with other battery cell layers in the place of lithium gel separator layer **130**. Once assembly of the battery cell's layers are complete and the battery cell has been inserted into a housing (e.g., a pouch), a liquid electrolyte may be added, then heat and pressure may be applied.

(31) At block **370** the liquid electrolyte mixture may be permeated into the non-reactive scaffold. Arrow **315** represents that lithium liquid is permeated throughout scaffolding material **305** to create lithium liquid-permeated scaffolding material **320**. The lithium liquid may permeate the scaffolding material when left submerged at atmospheric pressure for a duration of time, such as between 6 hours and 24 hours. This step may be performed after the dry separator layer has been assembled as part of a battery cell. The liquid electrolyte, which can be a lithium liquid, may be composed of materials that after permeating into the voids within the scaffolding material and then subjected to heat, causing the lithium liquid at least partially solidify, such as into a gel. Such an arrangement allows for a lithium gel separator layer to be initially created as a dry separator layer then for lithium liquid to be permeated throughout the scaffolding material and transitioned into being a gel after the battery cell is housed within a housing (e.g., a pouch cell). In some embodiments, pressure

is also applied, the purpose of the pressure may be to crease the amount of contact at various interfaces within the layered stack.

(32) After the lithium liquid has been permeated throughout scaffolding material **305** to create lithium liquid-permeated scaffolding material **320**, block **375** may be performed in which pressure, heat, or both may be applied, as indicated by arrow **325**. For instance, in some embodiments, pressure may first be applied at room temperature, at a force of 100 N/cm.^{sup.2} for a duration of between 60 and 120 seconds. A heat press process may then be performed at a temperature of between 100° C.-110° C. at a force of between 100-500 N/cm.^{sup.2} for a duration of between 60 and 120 seconds. The heat applied to lithium gel separator layer **301** may cause lithium liquid-permeated scaffolding material **320** to transition into pseudo-solid lithium gel layer **330**. In some embodiments, pressure may assist the process or may help increase the surface area of interfaces between layers of the lithium gel separator layer and/or other layers of the battery cell. Lithium gel separator layer **302** may then be finally formed.

(33) To create such a battery cell, various methods may be performed. FIG. **4** illustrates an embodiment of a method **400** for manufacturing a pouch-style battery cell that contains an anode-free solid-state battery having a lithium gel separator layer and an anti-dendrite layer. At block **405**, an anti-dendrite layer may be layered onto an anode current collector. The anti-dendrite layer may be as detailed in relation to anti-dendrite layer **140** of FIG. **1**. The anode current collector may be as detailed in relation to anode current collector **150** of FIG. **1**.

(34) At block **410**, a three-part lithium gel separator layer may be created. Initially the lithium gel separator layer may be in the form of a dry separator layer. That is, liquid electrolyte, such as lithium liquid, has not yet been injected into the scaffolding layer as detailed in relation to FIG. **3A**. In other embodiments, the scaffolding material at block **410** has been permeated with the liquid. The lithium gel separator layer, in which the lithium liquid may be present or not yet introduced, may be layered onto anti-dendrite layer at block **415**. The amount of adhesion between the lithium gel separator layer and the anti-dendrite layer may be greater than the amount of adhesion between the anti-dendrite layer and the anode current collector. In some instances, the amount of adhesion between the lithium gel separator layer and the anti-dendrite layer may be greater than the amount of adhesion between the anti-dendrite layer and the anode current collector after the heating and pressing of block **430**.

(35) At block **420** the cathode layer may be layered onto the lithium gel separator layer (of which the gel is still in the form of a liquid or has not yet been introduced to the scaffolding material). At block **425**, the cathode collector layer may be layered onto the cathode layer. In some embodiments, block **425** may be performed, then the combined cathode and cathode current collector layers may be layered onto the lithium gel separator layer (of which the gel is still in the form of a liquid or has not yet been introduced to the scaffolding material) at block **420**.

(36) It should be understood that blocks **405-425** may be repeated multiple times to create multiple layer stacks for a solid state battery cell. For instance, 16 sets of layers may be created in a stack set similar to detailed in relation to FIG. **2**. Such an arrangement allows for the anode current collector and cathode current collector to be in contact with anti-dendrite layers and cathodes, respectively, on opposite sides.

(37) At block **430**, the one or more layer stacks may be packaged in a pouch cell. At this block, the lithium liquid (or other form of liquid electrolyte), assuming it was introduced at block **410**, may still be in liquid form. The layers stacks may be vacuum sealed within the pouch cell to remove excess air. The pouch cell may be made of a flexible material, such as plastic, that can allow the pouch to expand and be compressed. If the lithium liquid was not permeated throughout the scaffolding layer at block **410**, the lithium liquid may be introduced within the pouch cell when packaging is being performed (or before or after) at block **432**. The lithium liquid may then permeate into the scaffolding layer of the dry separator layer.

(38) At block **435**, one or more processes of heat, pressure, or both may be applied to the packaged

pouch cell. This process can perform multiple functions: 1) block **435** may increase the amount of physical contact between adjacent layers of the battery cell; 2) block **435** may cause the lithium liquid to convert to a lithium gel; and 3) block **435** may create adhesion between the anti-dendrite layers and the lithium gel layers that is greater than the amount of adhesion between the anti-dendrite layers and the anode current collectors. For instance, in some embodiments, pressure may first be applied at room temperature, at a force of between 80 and 120 N/cm.² for a duration of between 60 and 120 seconds. A heat press process may then be performed at a temperature of between 80° C.-130° C. at a force of between 100-500 N/cm.² for a duration of between 60 and 120 seconds.

(39) At block **440**, the pouch cell may be installed within a jig press (or some other mechanical device that applies pressure to the pouch cell). The jig press may be used to apply long-term pressure to the SSB pouch cell. In some embodiments, multiple SSB pouch cells are layered and then compressed using the jig press. While in the jig press, the SSB pouch cells may be repeatedly charged and discharged. The SSB pouch cells may be used to power a vehicle or some other form of electrically-powered device.

(40) As an example of an SSB in accordance with FIG. 2 and that may be manufactured according to method **400**, an SSB pouch cell may be created that includes 16 layer sets and is approximately 65 mm by 65 mm. For a given layer, at 0% SOC (state of charge) a thickness of 3.3 mm is present. At 100% SOC, a thickness of 3.64 mm is present, representing approximately a 10% increase caused by swelling. The performance of the overall cell may be 4520 mAh with an average voltage of 3.79 V. At 100% SOC, the energy density (by volume) may be 1122 Wh/L; the energy density (by weight), may be 432 Wh/Kg.

(41) The above embodiments are focused on the creation of planar layers of battery cells. Such layers may be used in pouch-style battery cells. In other embodiments, such as those detailed in relation to FIGS. 5 and 6, cylindrical battery cells may be created. Such cylindrical battery cells can have the same layering as detailed in relation to FIGS. 1-3, however the process to create the cylindrical cells may differ. Further detail regarding such embodiments is provided in relation to FIGS. 5 and 6.

(42) FIG. 5 illustrates an embodiment of a cylindrical battery press system **500**. Cylindrical battery press system **500** can include: compression mechanism **510**; heating element **520**; buffer material **530**; cylindrical pouch battery cell (also referred to as “battery cell”) **540**; temperature sensor **550**; support structure **560**; and platform **570**. Embodiments of cylindrical battery press system and related systems are detailed in U.S. patent application Ser. No. 16/412,338, entitled “Isostatic Press Devices and Processes for Cylindrical Solid-State Batteries,” filed on May 14, 2019, the entire disclosure of which is hereby incorporated by reference for all purposes. Other embodiments related to a system to isotropically pressurize a cylindrical battery cell are detailed in U.S. patent application Ser. No. 16/217,010, entitled “Hydraulic Isostatic Press Processes for Solid-State Batteries”, filed on Dec. 11, 2018, the entire disclosure of which is also hereby incorporated by reference for all purposes.

(43) Compression mechanism **510** may be approximately cylindrical in shape and have a cross-section that is similar to a halo. A gap along the curved sidewall of compression mechanism **510** may be present. On either side of this gap is edge **511** and edge **512**. By edge **511** being moved toward edge **512**, the volume within compression mechanism **510** may be decreased. Therefore, when edge **511** is away from edge **512**, the volume within compression mechanism is larger, allowing buffer material and/or battery cell **540** to be installed. When edge **511** is toward edge **512**, the volume within compression mechanism **510** is smaller, thus applying pressure to buffer material **530** and, through buffer material **530**, to battery cell **540**.

(44) Battery cell **540** may be a cylindrical jelly-roll style battery cell, such as one similar to the embodiments of FIGS. 1-3. The cylindrical jelly-roll style battery cell may (initially) be stored inside of a pouch, which can be compressed using cylindrical battery press system **500**. After being

compressed and heated using cylindrical battery press system **500**, the cylindrical jelly-roll style battery cell may be removed from the pouch and installed within a cylindrical canister, such as detailed in method **600** of FIG. **6**.

(45) Compression mechanism **510** may be formed from a semi-rigid material, such as a hard rubber, plastic, or a layer of metal. Compression mechanism **510** may be partially deformed by edge **511** being pushed or pulled toward edge **512**. In some embodiments, edge **512** may be fixed to support structure **560**. Edge **511** may be connected with an extension, such as a metal bar, that allows a user to manually push or pull the metal bar to move edge **511** toward edge **512**. In other embodiments, a hydraulic pump or electric motor may be used to move edge **511** toward edge **512**.

(46) Buffer material **530** may be wrapped around battery cell **540**. Buffer material **530** may be a semi-rigid material, such as heat resistant rubber. In some embodiments, buffer material **530** may be a rubber or other form of flexible skin that is filled with liquid. Buffer material **530**, when viewed as a cross-section, may generally be halo-shaped. This halo shape defines a void within its center, into which a battery cell can be placed. Buffer material **530** may serve to transfer pressure applied by compression mechanism **510** to battery cell **540**. Buffer material **530** may help distribute the pressure applied by compression mechanism **510** such that the pressure applied to the curved sidewall of battery cell **540** is uniform or nearly uniform. In some embodiments, buffer material **530** is first wrapped around battery cell **540**. In some embodiments, buffer material **530** may be a sheet of buffer material in which battery cell is rolled. Therefore, the jelly-roll style battery cell may, in turn, be within a jelly-roll of buffer material. Buffer material **530** may be installed with compression mechanism **510**.

(47) Between buffer material **530** and compression mechanism **510**, heating element **520** may be present. Heating element **520** may be generally cylindrical in shape and may have a gap along the curved sidewall that matches the gap of compression mechanism **510**. Heating element **520** may be a resistive heater such that when current is applied to heating element **520**, heat is generated. In some embodiments, heating element **520** is capable of heating up to 250° C. The amount of heat output by heating element **520** may be controlled based on the output of temperature sensor **550**. Temperature sensor **550** may be located between battery cell **540** and buffer material **530**.

Therefore, temperature sensor **550** may indicate the temperature at an external surface of battery cell **540**. In some embodiments, it may be desirable for battery cell **540** to be heated to between 80° C. and 120° C. By applying a greater temperature using heating element **520**, it may be possible for battery cell **540** to be heated to between 80° C. and 120° C. at its surface quicker. An external heating controller (not pictured) may receive temperature measurements from temperature sensor **550** and control the amount of heat generated by heating element **520**.

(48) While edge **512** is fixed to support structure **560**, which is in turn fixed to platform **570**, edge **511** may remain free. By edge **511** remaining free from support structure **560** and platform **570**, edge **511** may be moved toward edge **512**, thus slightly deforming compression mechanism **510**. When force is ceased to be applied to edge **511**, compression mechanism **510** may expand back to a natural shape and pressure may cease being applied to battery cell **540**. It should be understood that the force applied to edge **511** may be applied in the vicinity of edge **511** and not necessarily precisely on edge **511**. However, the closer such force is applied to edge **511**, the more evenly distributed the pressure applied to buffer material **530** may be. Similarly, it should be understood that edge **512** can be directly fixed to support structure **560**, but rather a portion of compression mechanism **510** in a vicinity of edge **512** may be fixed to support structure **560**. Again here, the portion of compression mechanism **510** to edge **512** fixed to support structure **560**, the more evenly distributed the pressure applied to buffer material **530** may be.

(49) To create a cylindrical battery cell, various methods may be performed. FIGS. **6A** and **6B** illustrates an embodiment of a method for creating a cylindrical anti-dendrite anode-free solid-state battery. In block **605-620**, various steps may be performed to create a layered stack similar to as presented in and described in relation to FIG. **1**. In other embodiments, blocks **605-620** may be

performed to create a stack set as detailed in relation to FIG. 2. That is, blocks **605-620** may be performed multiple times to two or more (e.g., 3-20) sets of layers.

(50) Blocks **605-620** represent a possible embodiment of how multiple layers may be stacked together. In other embodiments, the ordering of blocks **605-620** may be different. At block **605**, a cathode layer may be attached to a cathode current collector layer. The cathode current collector layer may be aluminum foil and the cathode may, for example, be NCA (Nickel-Cobalt-Aluminum Oxide) or NCM (nickel-manganese-cobalt). The cathode layer may be deposited onto the cathode current collector layer or the cathode current collector layer may be deposited onto the cathode layer.

(51) At block **610**, an anti-dendrite layer may be deposited onto or otherwise attached with an anode current collector layer. The anode current collector layer may be copper and the anti-dendrite layer may be carbon black; acetylene back black; silver; zinc; gold; bismuth; tin; polyvinylidene fluoride (PvDF); polyimide (PI); polyacrylic acid (PAA); and carboxymethyl cellulose styrene-butadiene rubber (CMC-SBR). The anti-dendrite layer may also be formed using alloys of silver, zinc, gold, bismuth, and tin. The anti-dendrite layer may serve to decrease the nucleation energy for lithium ions to deposit onto a surface of the anode current collector layer.

(52) At block **615**, a dry separator layer, such as dry separator layer **300**, may be attached to either the cathode layer or the anti-dendrite layer or otherwise positioned between them. The dry separator layer can include two adhesive layers and a scaffolding material, such as detailed in relation to dry separator layer **300**. At block **620**, a layered stack may be created that includes the dry separator layer being layered between the cathode and the anti-dendrite layer. The layered stack may include at least: the dry separator layer, the anti-dendrite layer, the anode current collector layer, the cathode layer, and the cathode current collector layer, such as illustrated to FIG. 1 (with the dry separator layer in lieu of the lithium-gel separator layer).

(53) At block **625**, the layered stack may be rolled onto itself multiple times to create a jelly-roll style battery cell. When the layers are rolled together, a roughly cylindrical rolled layered stack can be created. At block **630**, the rolled layered stack may be inserted within a compressible, flexible pouch. The pouch may serve as a temporary housing for the battery cell during a portion of the manufacturing process. Prior to sealing the pouch, an electrolyte solution may be injected or added into the pouch. The electrolyte solution may be the lithium liquid may be as detailed in relation to FIG. 3A. The injection of the lithium liquid may cause the scaffolding material of the dry separator layer to be permeated by the lithium liquid and become lithium-liquid permeated scaffolding material, such as lithium-liquid permeated scaffolding material **320**. As part of block **635**, the pouch may have any air present removed and may be sealed.

(54) At block **640**, pressure may be applied to the pouch. The pressure may be applied using a system similar to cylindrical battery press system **500**. Prior to pressure being applied, a temperature probe may be inserted such that the temperature probe is adjacent to an external surface of the pouch within the cylindrical battery press system. Pressure may then be applied by the cylindrical battery press system either manually or using a motorized or hydraulic embodiment. The pressure applied may be between 100 kPa and 100 MPa. In some embodiments, pressure is applied for between 30 seconds and 1 hour.

(55) At block **645**, which may be performed at the same time as block **640** or at least partially overlapping in time with block **640**, heat may be applied. The amount of heat applied may be between 150° C. and 250° C. The temperature of the pouch may be monitored using the temperature probe. Heat may be applied until the battery cell is between 60° C. and 150° C. for a period of time, such as between 30 seconds and 1 hour. The pressure, heat, or both may cause the lithium liquid that permeates the scaffolding layer to transition to being a pseudo-solid lithium gel layer. Therefore, no liquid remains within the pouch. Accordingly, the battery cell is a solid-state battery cell (that includes a gel).

(56) The heat and pressure applied at blocks **640** and **645** may additionally or alternatively increase

the amount of surface area contact between one or more of the layers of the battery cell.

Additionally or alternatively, the heat and pressure may increase the adhesion among two or more of the layers of the battery cell.

(57) At block **650**, the cylindrical jelly roll that has been subjected to the heat and pressure may be removed from the pouch. No liquid may be presented because it has transitioned into a gel within the scaffolding layer. The cylindrical jelly roll may be inserted into a cylindrical battery cell canister. The cylindrical battery cell canister may be rigid or semi-rigid. In some embodiments, the cylindrical battery cell canister may be metal. The cylindrical battery cell canister may exert pressure on the cylindrical jelly roll when the cylindrical jelly roll expands. For example, when the battery cell is charged at block **660**, lithium deposition on the anode current collector can cause the battery to swell 0.5% to 3% in diameter. Pressure exerted by the sidewalls of the cylindrical battery cell canister can help control the amount of swelling and help keep layers of the battery cell in contact with each other. At block **660**, the battery cell may repeatedly be charged and discharged to power an electric device, such as an electric vehicle (EV). A cylindrical battery manufactured according to method **600** may be charged to 7559 Ah, may discharge 6229 Ah, thus exhibiting an initial Columbic efficiency of 82.4%.

(58) When lithium deposits on an anode-current collector during charging, it may tend to deposit in pools, rather than in a roughly flat film. Since the amount of contact present between the deposited lithium and the anode current collector can be small, the electrical connection between the deposited lithium and the anode current collector may be small. Having a small or weak electrical connection between the deposited lithium and the anode current collector can cause the impedance of the battery cell to be high. A high impedance can result in reduced performance of the battery cell: that is, a battery with a low internal resistance may be able to deliver a large amount of current on demand. For some applications, like use in an electric vehicle (EV), the ability to deliver current quickly can greatly affect performance, such as the ability of the EV to accelerate. When a battery cell has a high internal resistance, current flowing through the battery cell can cause the battery to heat up, which can damage the battery cell.

(59) In some embodiments, an additional layer may be present within the layer stack of an anode-free solid-state battery. The additional layer may be situated between the anti-dendrite layer and the anode current collector. This layer can be referred to as an interfacial bonding layer. The interfacial bonding layer may encourage formation of lithium deposits with a high degree of surface contact between the interfacial bonding layer and the lithium deposits. Since the interfacial bonding layer has a large amount of contact with both the anode current collector and deposited lithium, the internal resistance of the battery cell can be decreased. Such an interfacial bonding layer may be added to any of the embodiments detailed in relation to FIGS. **1-6** or as detailed in relation to FIGS. **7-10**.

(60) An interfacial layer may be made from conductive agents and binder. In some embodiments, the interfacial layer may be between 30%-99% conductive agent; the remainder of the interfacial bonding layer may be binder (1%-70%). FIG. **7** illustrates an embodiment of a layer stack **700** of an anode-free solid-state battery having a lithium gel separator layer, an anti-dendrite layer, and an interfacial bonding layer. Layer stack **700** may be as detailed in relation to FIG. **1**, however interfacial bonding layer **710** may be presented between anode current collector **150** and anti-dendrite layer **140**. Interfacial bonding layer **710** may be in direct contact on a first side with anti-dendrite layer **140** and in direct contact on a second, opposite side with anode current collector **150**.

(61) In order to encourage deposition of lithium metal on interfacial bonding layer **710**, interfacial bonding layer **710** may have a first amount of adhesion with anode current collector **150** that is greater than a second amount of adhesion between anti-dendrite layer **140** and interfacial bonding layer **710** or a third amount of adhesion between anti-dendrite layer **140** and Lithium gel separator layer **130**. The thickness of interfacial bonding layer **710** may be between 0.05 μm and 5 μm . The density of the interfacial bonding layer may be between 0.1 and 2.0 grams per cubic centimeter.

(62) Interfacial bonding layer **710** may use carbon as a conductive agent mixed with binders (PvDF, SBR-CMC, PAA) and metal particles such as Bi, Sn, Ag, Au, Pt. More specifically, acetylene black or carbon black may be used as the conductive agent. The individual carbon particles may be between 3 nm and 20 nm spherical shape particles. Possible types of bonders include: PvDF, SBR-CMC, and PAA.

(63) The impedance or resistance of the battery cell of FIG. 7, as measured between terminal **720** and terminal **730** may be significantly decreased by the presence of interfacial bonding layer **710** as compared to an embodiment, such as FIG. 1, where interfacial bonding layer **710** is not present. As an example, an embodiment of a battery cell without an interfacial bonding layer may have an impedance of 0.85 ohms; but if an interfacial bonding layer is present between the anode current collector and the anti-dendrite layer, the impedance may be 0.05 ohms.

(64) FIG. 8 illustrates an embodiment of a layer stack **800** of an anode-free solid-state battery that indicates relative amounts of adhesion between various layers. One of the key aspects of layer stack **700** of FIG. 7 may be that the relative amount of adhesion between the layers encourages lithium metal to plate during the charging process between anti-dendrite layer **140** and interfacial bonding layer **710**.

(65) Interface **801** between lithium gel separator layer **130** and anti-dendrite layer **140** may have a first amount of adhesion. Interface **802** between anti-dendrite layer **140** and interfacial bonding layer **710** may have a second amount of adhesion. Interface **803** between interfacial bonding layer **710** and anode current collector **150** may have a third amount of adhesion. By interface **802** having less adhesion than interface **801** or interface **803**, lithium may be encouraged to plate at interface **802**. Stated another way, the second amount of adhesion may be greater than the first amount of adhesion or the third amount of adhesion.

(66) FIG. 9 illustrates an embodiment **900** of a layer stack of an anode-free solid-state in which lithium ions migrate and are deposited onto the interfacial bonding layer. In embodiment **900**, the battery cell is being charged. Charging causes lithium ions to migrate from cathode **120**, through lithium gel separator layer **130**, through anti-dendrite layer **140** and plate as lithium metal layer **910** between anti-dendrite layer **140** and interfacial bonding layer **710**, as indicated by arrows **905**. Anti-dendrite layer **140** may help inhibit the growth of dendrites that could pierce lithium gel separator layer **130**. Therefore, interfacial bonding layer **710** may be used in conjunction with anti-dendrite layer **140**.

(67) The presence of lithium metal layer **910** may cause swelling in the battery cell. During a discharge cycle, lithium ion may migrate from lithium metal layer **910** to cathode **120**. Swelling within the battery cell may decrease as the battery cell is discharged and lithium ions migrate to cathode layer **120**.

(68) FIG. 10 illustrates an embodiment of a method **1000** for manufacturing a pouch-style battery cell that contains a solid-state battery having a lithium gel separator layer, an anti-dendrite layer, and an interfacial bonding layer. It should be understood that method **1000** can be adapted in accordance with the blocks of method **600** of FIGS. 6A and 6B such that a interfacial bonding layer can be manufactured as part of a cylindrical battery cell.

(69) At block **1005**, an interfacial bonding layer can be deposited onto the anode current collector. The anode current collector may be as detailed in relation to anode current collector **150** of FIG. 1. Block **1005** can include a conductive material, such as acetylene black, being mixed with a bonder and deposited onto the anode current collector.

(70) At block **1010**, a three-part lithium gel separator layer may be created. Initially the lithium gel separator layer may be in the form of a dry separator layer. That is, liquid electrolyte, such as lithium liquid, has not yet been injected into or permeated throughout the scaffolding layer as detailed in relation to FIG. 3A. In other embodiments, the scaffolding material has been permeated with the liquid. An anti-dendrite layer may be layered onto the lithium gel separator layer, in which the lithium liquid (or another liquid electrolyte) may be present or not yet introduced at block **1020**.

The anti-dendrite layer may be as detailed in relation to anti-dendrite layer **140** of FIG. **1**. In some instances, the amount of adhesion between the lithium gel separator layer and the anti-dendrite layer may be greater than the amount of adhesion between the anti-dendrite layer and the interfacial bonding layer.

(71) At block **1025** the cathode layer may be layered onto the lithium gel separator layer (of which the gel is either still in the form of a liquid or not yet present). At block **1030**, the cathode collector layer may be layered onto the cathode layer. In some embodiments, block **1030** may be performed, then the combined cathode and cathode current collector layers may be layered onto the lithium gel separator layer at block **1025**.

(72) At block **1032**, the anti-dendrite layer that was previously layered onto the lithium-gel separator layer may have its opposite side layers layered onto the interfacial bonding layer. The layering of the anti-dendrite layer and the interfacial bonding layer may result in relatively little adhesion being present between the layers. The anti-dendrite layer may create an interface that has less adhesion with the interfacial bonding layer than the interfacial bonding layer forms with the anode current collector. The amount of adhesion can be controlled by modulating the binder and active conductive material ratio of the interfacial bonding layer. For example, PvDF may be used as the binder and kitchen black may be used as the active material in a ratio of 3% PvDF to 97% kitchen black. In other embodiments, the ration of kitchen black is between 95% and 98%.

(73) It should be understood that blocks **1005-1030** may be repeated multiple times to create multiple layer stacks for a solid state battery cell. For instance, 16 sets of layers may be created in a stack set similar to detailed in relation to FIG. **2** with the addition of interfacial bonding layers. Such an arrangement allows for the anode current collector and cathode current collector to be in contact with anti-dendrite layers and cathodes, respectively, on opposite sides.

(74) At block **1035**, the one or more layer stacks may be packaged in a pouch cell. At this block, the lithium liquid (or other form of liquid electrolyte), assuming it was introduced at block **1015**, may still be in liquid form. The layers stacks may be vacuum sealed within the pouch cell to remove excess air. The pouch cell may be made of a flexible material, such as plastic, that can allow the pouch to expand and be compressed. If the liquid electrolyte, such as lithium liquid was not permeated throughout the scaffolding layer at block **1015**, the liquid electrolyte may be introduced within the pouch cell when packaging is being performed (or before or after) at block **1040**. The lithium liquid may then permeate into the scaffolding layer of the dry separator layer.

(75) At block **1045**, one or more processes of heat, pressure, or both may be applied to the packaged pouch cell. This process can perform multiple functions: 1) block **1045** may increase the amount of physical contact between adjacent layers of the battery cell; 2) block **1045** may cause the liquid electrolyte (e.g., lithium liquid) to convert to a lithium gel; and 3) block **1045** can help create adhesion between the interfacial bonding layers and the anode current collectors that is greater than the amount of adhesion between the anti-dendrite layers and the interfacial bonding layers. For instance, in some embodiments, pressure may first be applied at room temperature, at a force of between 80 and 120 N/cm.^{sup.2} for a duration of between 60 and 120 seconds. This portion of the process may increase the amount of contact present at one of more interfaces of the layers of the battery cell. A heat press process may then be performed at a temperature of between 50° C.-130° C. at a force of between 100-1000 N/cm.^{sup.2} for a duration of between 60 and 2400 seconds.

(76) At block **1050**, the pouch cell may be installed within a jig press (or some other mechanical device that applies pressure to the pouch cell). The jig press may be used to apply long-term pressure to the SSB pouch cell. In some embodiments, multiple SSB pouch cells are layered and then compressed using the jig press. While in the jig press, the SSB pouch cells may be repeated charged and discharged. The SSB pouch cells may be used to power a vehicle or some other form of electrically-powered device.

(77) The methods, systems, and devices discussed above are examples. Various configurations may omit, substitute, or add various procedures or components as appropriate. For instance, in

alternative configurations, the methods may be performed in an order different from that described, and/or various stages may be added, omitted, and/or combined. Also, features described with respect to certain configurations may be combined in various other configurations. Different aspects and elements of the configurations may be combined in a similar manner. Also, technology evolves and, thus, many of the elements are examples and do not limit the scope of the disclosure or claims.

(78) Specific details are given in the description to provide a thorough understanding of example configurations (including implementations). However, configurations may be practiced without these specific details. For example, well-known processes, structures, and techniques have been shown without unnecessary detail in order to avoid obscuring the configurations. This description provides example configurations only, and does not limit the scope, applicability, or configurations of the claims. Rather, the preceding description of the configurations will provide those skilled in the art with an enabling description for implementing described techniques. Various changes may be made in the function and arrangement of elements without departing from the spirit or scope of the disclosure.

(79) Also, configurations may be described as a process which is depicted as a flow diagram or block diagram. Although each may describe the operations as a sequential process, many of the operations can be performed in parallel or concurrently. In addition, the order of the operations may be rearranged. A process may have additional steps not included in the figure.

(80) Having described several example configurations, various modifications, alternative constructions, and equivalents may be used without departing from the spirit of the disclosure. For example, the above elements may be components of a larger system, wherein other rules may take precedence over or otherwise modify the application of the invention. Also, a number of steps may be undertaken before, during, or after the above elements are considered.

Claims

1. An anode-free solid-state battery, comprising: a cathode layer comprising lithium ions; an anode current collector layer comprising copper; a lithium-gel separator layer between the cathode layer and the anode current collector layer, wherein the lithium-gel separator layer comprises: a scaffolding material having spaces within the scaffolding material; and a lithium gel formed from a liquid electrolyte filled in the spaces and comprising a salt, a solvent, and a cross-linked polymer of a polymer additive and a cross-linker additive; an anti-dendrite layer located between the lithium-gel separator layer and the anode current collector layer, wherein the anti-dendrite layer comprises one or more materials selected from the group consisting of: carbon black; acetylene black; ketchen black; silver; zinc; gold; bismuth; tin; polyimide (PI); polyacrylic acid (PAA); and carboxymethyl cellulose styrene-butadiene rubber (CMC-SBR), the anti-dendrite layer discourages dendrite formation; and an interfacial bonding layer located between the anti-dendrite layer and the anode current collector layer, wherein: the interfacial bonding layer is in direct contact on a first side with the anode current collector layer; the interfacial bonding layer consists of a conductive agent, metal particles, and a binder material, wherein the binder material is selected from the group consisting of polyvinylidene fluoride (PvDF), styrene butadiene rubber and sodium carboxyl methyl cellulose (SBR/CMC), poly(acrylic acid) (PAA), and any combinations thereof, the conductive agent is selected from the group consisting of acetylene black, carbon black, and a combination thereof, the metal particles are selected from the group consisting of Bi particles, Sn particles, Ag particles, Au particles, Pt particles, and any combinations thereof, the interfacial bonding layer is free from lithium and encourages formation of lithium deposits between the interfacial bonding layer and the anti-dendrite layer; the interfacial bonding layer increases an amount of electrical connectivity between the anode current collector layer and lithium deposited on the interfacial bonding layer during charging of the anode-free solid-state battery; a first amount of adhesion between the

interfacial bonding layer and the anode current collector layer is greater than a second amount of adhesion between the anti-dendrite layer and the interfacial bonding layer; and a lithium metal layer plates between the anti-dendrite layer and the interfacial bonding layer when the anode-free solid-state battery is charging.

2. The anode-free solid-state battery of claim 1, wherein the anti-dendrite layer is formed of an alloy of one or more metals selected from the group consisting of: silver; zinc; gold; bismuth; and tin.

3. The anode-free solid-state battery of claim 1, wherein the scaffolding material has a spider-web structure.

4. The anode-free solid-state battery of claim 1, wherein: the salt, the polymer additive, and the cross-linker additive are dissolved into the solvent to form the liquid electrolyte; the liquid electrolyte is dispersed in the spaces throughout the scaffolding material; and the liquid electrolyte, after being dispersed throughout the scaffolding material is converted to the lithium gel.

5. The anode-free solid-state battery of claim 1, wherein: the interfacial bonding layer is between 0.05 μm and 5 μm ; and the anti-dendrite layer is between 0.05 micrometers and 10 micrometers in thickness.

6. A method for creating an anode-free solid-state battery, the method comprising: depositing an interfacial bonding layer onto an anode current collector layer; creating a lithium-gel separator layer; depositing an anti-dendrite layer onto the lithium-gel separator layer; layering a cathode and cathode current collector onto the lithium-gel separator layering the anti-dendrite layer onto the interfacial bonding layer, wherein: a first amount of adhesion between the anti-dendrite layer and the interfacial bonding layer is less than a second amount of adhesion between the interfacial bonding layer and the anode current collector layer; and the interfacial bonding layer increases an amount of electrical connectivity between the anode current collector layer and lithium deposited on the interfacial bonding layer; and wherein the anode current collector layer comprises copper; wherein the cathode layer comprises lithium ions; wherein the anti-dendrite layer comprises one or more materials selected from the group consisting of: carbon black; acetylene black; ketchen black; silver; zinc; gold; bismuth; tin; polyimide (PI); polyacrylic acid (PAA); and carboxymethyl cellulose styrene-butadiene rubber (CMC-SBR); wherein the lithium-gel separator layer comprises: a scaffolding material having spaces within the scaffolding material; and a lithium gel formed from a liquid electrolyte filled in the spaces and comprising a salt, a solvent, and across-linked polymer of a polymer additive and a cross-linker additive; wherein the interfacial bonding layer: is in direct contact on a first side with the anode current collector layer; consists of a conductive agent, metal particles, and a binder material, wherein the binder material is selected from the group consisting of polyvinylidene fluoride (PvDF), styrene butadiene rubber and sodium carboxyl methyl cellulose (SBR/CMC), poly(acrylic acid) (PAA), and any combinations thereof, the conductive agent is selected from the group consisting of acetylene black, carbon black, and a combination thereof, the metal particles are selected from the group consisting of Bi particles, Sn particles, Ag particles, Au particles, Pt particles, and any combinations thereof, the interfacial bonding layer is free from lithium and encourages formation of lithium deposits between the interfacial bonding layer and the anti-dendrite layer; and a lithium metal layer plates between the anti-dendrite layer and the interfacial bonding layer when the anode-free solid-state battery is charging.

7. The method for creating the anode-free solid-state battery of claim 6, wherein a presence of the interfacial bonding layer decreases an impedance of the anode-free solid-state battery.

8. The method for creating the anode-free solid-state battery of claim 6, wherein creating the lithium-gel separator layer comprises: attaching a first adhesive layer to scaffolding material wherein the scaffolding material defines voids within the scaffolding material; attaching a second adhesive layer to the scaffolding material; and permeating the liquid electrolyte into the spaces within the scaffolding material.

9. The method for creating the anode-free solid-state battery of claim 8, wherein the anti-dendrite

layer is deposited onto the lithium-gel separator layer prior to the liquid electrolyte being permeated into the spaces within the scaffolding material.

10. The method for creating the anode-free solid-state battery of claim 8, wherein creating the lithium-gel separator layer further comprises: after permeating the liquid electrolyte into the spaces within the scaffolding material, applying heat to the lithium-gel separator layer that causes the liquid electrolyte to become the lithium gel.

11. The method for creating the anode-free solid-state battery of claim 6, wherein the anti-dendrite layer consists of an alloy of one or more metals selected from the group consisting of: silver; zinc; gold; bismuth; and tin.

12. A pouch-style anode-free solid-state battery, comprising: a flexible pouch; a cathode layer comprising lithium ions; an anode current collector layer comprising copper; a lithium-gel separator layer between the cathode layer and the anode current collector layer, wherein the lithium-gel separator layer comprises: a scaffolding material having spaces within the scaffolding material; and a lithium gel formed from a liquid electrolyte filled in the spaces and comprising a salt, a solvent, and a cross-linked polymer of a polymer additive and a cross-linker additive; an anti-dendrite layer located between the lithium-gel separator layer and the anode current collector layer, wherein the anti-dendrite layer comprises one or more materials selected from the group consisting of: carbon black; acetylene black; ketchen black; silver; zinc; gold; bismuth; tin; polyimide (PI); polyacrylic acid (PAA); and carboxymethyl cellulose styrene-butadiene rubber (CMC-SBR), the anti-dendrite layer discourages dendrite formation; and an interfacial bonding layer located between the anti-dendrite layer and the anode current collector layer, wherein: the interfacial bonding layer is in direct contact on a first side with the anode current collector layer; the interfacial bonding layer consists of a conductive agent, metal particles, and a binder material, wherein the binder material is selected from the group consisting of polyvinylidene fluoride (PvDF), styrene butadiene rubber and sodium carboxyl methyl cellulose (SBR/CMC), poly(acrylic acid) (PAA), and any combinations thereof, the conductive agent is selected from the group consisting of acetylene black, carbon black, and a combination thereof, the metal particles are selected from the group consisting of Bi particles, Sn particles, Ag particles, Au particles, Pt particles, and any combinations thereof, the interfacial bonding layer is free from lithium and encourages formation of lithium deposits between the interfacial bonding layer and the anti-dendrite layer; the interfacial bonding layer increases an amount of electrical connectivity between the anode current collector layer and lithium deposited on the interfacial bonding layer; a first amount of adhesion between the interfacial bonding layer and the anode current collector layer is greater than a second amount of adhesion between the anti-dendrite layer and the interfacial bonding layer; the cathode layer, the anode current collector layer, the lithium-gel separator layer, the anti-dendrite layer, and the interfacial bonding layer are inserted within the flexible pouch; and a lithium metal layer plates between the anti-dendrite layer and the interfacial bonding layer when the anode-free solid-state battery is charging.

13. The pouch-style anode-free solid-state battery of claim 12, wherein the conductive agent is in a form of carbon particles with an average size from 3 nm to 20 nm.

14. The anode-free solid-state battery of claim 1, wherein the binder material is selected from the group consisting of styrene butadiene rubber and sodium carboxyl methyl cellulose (SBR/CMC), poly(acrylic acid) (PAA), and a combination thereof.

15. The pouch-style anode-free solid-state battery of claim 12, wherein the binder material is selected from the group consisting of styrene butadiene rubber and sodium carboxyl methyl cellulose (SBR/CMC), poly(acrylic acid) (PAA), and a combination thereof.

16. The anode-free solid-state battery of claim 1, wherein the lithium-gel separator layer functions as a solid electrolyte; and the scaffolding material has a porosity of between 10% and 90%.

17. The anode-free solid-state battery of claim 1, wherein the polymer additive comprises Poly (4-vinylpyridine) (P4VP); and the cross-linker additive is selected from the group consisting of

N,N,N',N'-Tetra(trifluoromethanesulfonyl)-dodecane-1,12-diamine (C12TFSA), N,N,N',N'-Tetra(trifluoromethanesulfonyl)-hexane-1,6-diamine (C6TFSA), and a combination thereof.

18. The pouch-style anode-free solid-state battery of claim 12, wherein the lithium-gel separator layer functions as a solid electrolyte; and the scaffolding material has a porosity of between 10% and 90%.

19. The pouch-style anode-free solid-state battery of claim 12, wherein the anti-dendrite layer consists of an alloy of one or more metals selected from the group consisting of silver; zinc; gold; bismuth; and tin.

20. The pouch-style anode-free solid-state battery of claim 12, wherein the polymer additive comprises Poly (4-vinylpyridine) (P4VP); and the cross-linker additive is selected from the group consisting of N,N,N',N'-Tetra(trifluoromethanesulfonyl)-dodecane-1,12-diamine (C12TFSA), N,N,N',N'-Tetra(trifluoromethanesulfonyl)-hexane-1,6-diamine (C6TFSA), and a combination thereof.

21. The anode-free solid-state battery of claim 1, wherein the binder material is styrene butadiene rubber and sodium carboxyl methyl cellulose (SBR/CMC).

22. The pouch-style anode-free solid-state battery of claim 12, wherein the binder material is styrene butadiene rubber and sodium carboxyl methyl cellulose (SBR/CMC).
